

Homecare Hospital Management System

Pharmacy Inventory, Patient Encounter Tracking & POS Billing Platform | Osogbo, Nigeria

CLIENT	LOCATION	HOSTING & BACKEND	PRINTER SPEC
Homecare Clinic	Osogbo, Osun State	Vercel + Supabase	58mm / 80mm Thermal POS

1. EXECUTIVE SUMMARY & CORE ARCHITECTURAL INTENT

Homecare Hospital requires a purpose-built web application to govern patient triage, nursing administration, pharmacy dispensary, and cashier reconciliations. The core differentiator of this system is its **decoupled billing model**: while the hospital maintains strict real-time auditability over pharmaceutical usage and inventory costs, patient receipts present a consolidated, treatment-based fee with no drug line items visible to clients.

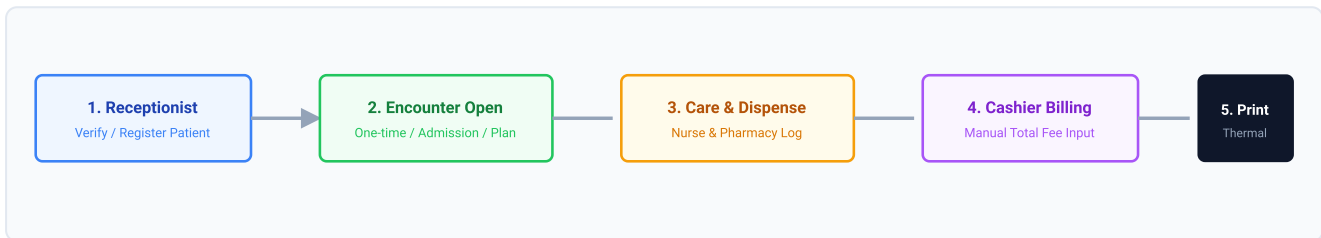
Guiding Architectural Rule

Under no circumstances must unit drug prices, brand names of dispensed medication, or injection costs appear on the customer-facing receipt or in the cashier's checkout view. Medication expenses remain invisible, absorbed directly into the manual treatment charge.

2. ROLE-BASED ACCESS CONTROL (RBAC) MATRIX

USER ROLE	PRIMARY RESPONSIBILITY	ACCESSIBLE FEATURES	STRICT RESTRICTIONS
Receptionist	Patient intake & triage encounter assignment	Register patient, check status, initiate one-time/admission/recurring treatment	No billing inputs, no pharmacy stock adjustments
Nurse	Patient care notes & injection administration	Log vitals, record administered injections & ward dosages to active treatment	No payment handling, no access to purchase costs
Pharmacy	Inventory control & drug dispensary	Stock entry, reorder thresholds, batch management, dispensing to encounters	Cannot issue receipts or post monetary payments
Cashier	Manual billing, partial tracking & receipting	Set treatment charges, record payments (cash/POS), issue thermal receipt	No access to drug cost or dispensary logs
Admin	Governance, analytics & reconciliations	Full system override, internal profit reports, user creation, audit logs	None

3. OPERATIONAL FLOW DIAGRAM



4. ENCOUNTER TYPES & LIFECYCLE STATE MACHINE

Every visit begins as a Treatment Encounter initiated by the receptionist. Treatment encounters maintain lifecycle states that control when pharmacy dispensations and cashier billing transactions can occur.

ENCOUNTER TYPE	OPERATIONAL DESCRIPTION	CLOSURE & SETTLEMENT CRITERIA
One-Time Treatment	Standard outpatient visit, emergency first-aid, or rapid clinical checkup.	Closed immediately upon 100% full balance payment. Single-session encounter.
Admission	Inpatient ward stays spanning multiple shifts, days, or weeks.	Permits rolling pharmacy dispensations until medical discharge. Payments logged iteratively.
Recurring Care	Antenatal, ulcer dressings, hypertension management, or chronic follow-ups.	Encounter stays open across multiple visits. Running balance ledger tracked per session.

5. SUPABASE DATABASE SCHEMA DESIGN

The relational data model below enforces data integrity and isolates financial privacy using PostgreSQL Row-Level Security (RLS) rules.

TABLE: patients

Encounter Base

id: uuid (PK, default gen_random_uuid())
patient_code: text UNIQUE (Format: HC-YYYY-XXXX)
full_name: text NOT NULL | phone: text | gender: text
registered_by: uuid (FK -> auth.users) | created_at: timestamptz (default now())

TABLE: treatments

Encounter Master

id: uuid (PK, default gen_random_uuid())
patient_id: uuid (FK -> patients.id) | encounter_type: encounter_enum (one_time, admission, recurring)
category: text (e.g., General Outpatient, Antenatal, Wound Care, Minor Surgery)
status: status_enum (active, discharged, completed, cancelled)
total_treatment_fee: numeric(12,2) DEFAULT 0.00 (Set manually by Cashier)
created_by: uuid (FK -> auth.users) | created_at: timestamptz

TABLE: pharmacy_inventory

Internal Stock Control

id: uuid (PK, default gen_random_uuid())
item_name: text NOT NULL | item_type: text (tablet, syrup, injection, consumable)
stock_quantity: integer DEFAULT 0 | reorder_level: integer DEFAULT 10
unit_cost_price: numeric(10,2) NOT NULL (Internal purchase rate)
updated_at: timestamptz

TABLE: treatment_dispensations

Internal Audit Ledger (Hidden from Cashier/Client)

id: uuid (PK, default gen_random_uuid())
treatment_id: uuid (FK -> treatments.id) | item_id: uuid (FK -> pharmacy_inventory.id)
quantity: integer NOT NULL | unit_cost_snapshot: numeric(10,2) NOT NULL
dispensed_by: uuid (FK -> auth.users) | dispensed_at: timestamptz

TABLE: payments

Cashier Transaction Ledger

id: uuid (PK, default gen_random_uuid())
receipt_number: text UNIQUE (Format: REC-YYYYMMDD-XXXX)
treatment_id: uuid (FK -> treatments.id) | amount_paid: numeric(12,2) NOT NULL
payment_method: payment_enum (cash, pos_terminal, bank_transfer)
cashier_id: uuid (FK -> auth.users) | created_at: timestamptz

6. PAYMENT MATH & LEDGER CALCULATIONS

The financial logic is decoupled to allow flexible clinical pricing while sustaining strict stock tracking.

- **Total Billed Amount** (`total_treatment_fee`): Manually defined by the cashier or medical officer based on clinical category, complexity, and internal discretion.
- **Cumulative Payments**: Sum of all successful transactions linked to `treatment_id`.
$$\text{Total_Paid} = \sum (\text{payments.amount_paid})$$
- **Remaining Balance**: Computed on every ledger query:
$$\text{Balance_Remaining} = \text{total_treatment_fee} - \text{Total_Paid}$$
- **Payment Status Flag**:
 - If `Total_Paid = 0` → **UNPAID**
 - If $0 < \text{Total_Paid} < \text{total_treatment_fee}$ → **PARTIAL**
 - If $\text{Total_Paid} \geq \text{total_treatment_fee}$ → **PAID IN FULL**
- **Encounter Completion**: An encounter status may only be transitioned to *Completed* if `Balance_Remaining ≤ 0`.

7. THERMAL POS RECEIPT LAYOUT (58MM / 80MM ESC-POS)

The thermal slip prints directly from the web app using standard CSS `@media print` rules or direct ESC/POS Bluetooth/USB bridge. Notice that **no medication breakdown exists**.

HEMOCARE CLINIC
Medical & Maternity Services
Osogbo, Osun State • Tel: 080-HEMOCARE

Date: 13-Sep-2026 10:45 AM
Receipt #: REC-20260913-0492
Patient ID: HC-2026-0041
Patient: [Patient Full Name]
Cashier: A. Adeleke

TREATMENT / SERVICE	AMOUNT
Outpatient Care Plan (Includes clinical care & nursing services)	₦ 18,500.00

Total Bill: ₦ 18,500.00
Amount Paid (POS): ₦ 10,000.00
Remaining Balance: ₦ 8,500.00
Payment Status: **PARTIAL**

Thank You & Quick Recovery
Retain receipt for subsequent visits

8. DEPLOYMENT & DELIVERY CHECKLIST FOR DEVELOPER

- **Frontend / Hosting**: Next.js App Router deployed on Vercel with responsive desktop/tablet layouts.
- **Database & Auth**: Supabase PostgreSQL instance with Email/Password auth and RBAC JWT claims.
- **Thermal Printing**: Calibrated CSS print stylesheet with `@page { size: 80mm auto; margin: 0; }` to eliminate browser headers and footers during POS output.